

Computational Finance

Lecture 11 – Exotic Options

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Lecture 11 – Exotic Options

- Why exotic options exist
- Before and after the 2008 crisis
- Exotic options still actively traded today
 - Barrier options¹
 - Asian options
 - Digital/binary options
 - Basket and worst-of options
 - Cliquet options
- Numerical method calculation procedures
- Conclusions

¹Covered in detail in Lecture 10; not revisited here.

Why Exotic Options? — Custom Payoffs and Cost-Efficient Hedging

- Vanilla options are often insufficient for real-world financial needs. Exotic options provide:
- Customized payoff structures.
 - *Example – FX digital hedge.* A European exporter wants protection only in a specific tail: pay a fixed amount if $\text{EUR/USD} > 1.10$ at maturity, nothing otherwise. The discontinuous payoff cannot be replicated cheaply with vanilla calls.
- Reduction of hedging costs.
 - *Example – Asian fuel hedge.* An airline hedging one month of jet-fuel procurement buys a single Asian call on the monthly average instead of ~ 20 daily vanilla calls. Averaging reduces effective volatility, so the premium is materially lower.
- Better matching of business exposures.
 - *Example – same airline.* The economic exposure is the average procurement cost $\frac{1}{n} \sum_{i=1}^n S_{t_i}$, not the terminal price S_T . An Asian option matches this exposure exactly; a vanilla terminal-price call would over- or under-hedge most of the month.

Why Exotic Options? — Yield, Structured Products, Vol/Correlation

- Yield enhancement in low-rate environments.

- *Example – worst-of autocallable.* On a basket of EURO STOXX 50, S&P 500 and Nikkei 225: pays a 10% annual coupon if the worst performer is above the barrier, with potential early redemption. The investor implicitly sells correlation and downside risk on the weakest index in return for a coupon well above prevailing rates.

- Structured investment products.

- *Example – capital-protected cliquet note.* Principal guaranteed at maturity; annual equity gains locked in subject to an 8% annual cap and 0% floor. A smooth, behaviourally appealing return profile – a workhorse of insurance and wealth-management offerings.

- Volatility and correlation trading.

- *Example – multi-asset structures and cliquets.* A worst-of put gives clean exposure to the correlation risk premium (the bank issuing it is short correlation); a cliquet gives forward-skew / forward-volatility exposure. These are first-order exposures that vanilla-option portfolios cannot replicate cleanly.

Who Uses Exotic Options?

- Main users:

- corporates – FX, commodity, and rate hedging;
- asset managers – tactical and thematic exposures;
- insurance companies – cliquet-style equity-linked guarantees;
- private banks – yield-enhancement notes: clients receive above-market coupons or income, funded by implicitly selling option-like risks (downside exposure, volatility, correlation);
- structured-product desks – autocallables, reverse convertibles;
- commodity and FX hedgers – Asian, barrier and digital products.

Exotic Options Before 2008

- Characteristics of pre-2008 markets:
 - cheap funding;
 - weak regulatory capital constraints;
 - large OTC derivative growth;
 - aggressive yield-enhancement demand;
 - heavy structured-product issuance.
- Popular structures before 2008:
 - barrier options;
 - reverse convertibles;
 - callable range accruals;
 - snowballs and autocallables;
 - long-dated path-dependent structures;
 - highly bespoke OTC exotics.
- Key theme: banks optimized yield and leverage more aggressively.

The 2008 Credit Crisis and Market Shift

- Major lessons from 2008:
 - counterparty risk became critical;
 - liquidity assumptions failed;
 - correlation structures broke down;
 - model risk became more visible;
 - funding costs became important.
- Post-2008 changes:
 - XVA adjustments (CVA, FVA, MVA);
 - Basel III capital constraints;
 - increased collateralization;
 - simplification of product books;
 - preference for hedgeable flow products.

Which Exotic Products Survived?

- Evolution of the exotic-product mix:

Category	Before 2008	After 2008
Highly bespoke OTC exotics	Very common	Reduced
Barrier products	Very active	Still active
Asian options	Active	Active
Digitals	Active	Active
Basket/Worst-of	Growing	Very active
Cliquets	Active (aggressive variants)	Reduced (insurance-driven)
Long-dated exotics	Common	Reduced
Autocallables	Growing	Dominant in equity SPs

- **Main trend:** modern markets prefer products that are
 - easier to hedge;
 - easier to model;
 - easier to explain to regulators;
 - more liquid.

Asian Options: Definition

- Asian options depend on the **average price** of the underlying.
- *Arithmetic average price call*:

$$\max\left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0\right).$$

- *Geometric average price call*:

$$\max\left(\left(\prod_{i=1}^n S_{t_i}\right)^{1/n} - K, 0\right).$$

- Key properties:
 - path dependent;
 - reduced volatility exposure;
 - cheaper than vanilla options;
 - popular in commodities and FX.

Asian Options: Real-World Example

- **Airline fuel hedging.** Suppose an airline buys jet fuel continuously over one month.
- A vanilla option only protects the terminal price S_T , but the airline's exposure is closer to

$$\frac{1}{n} \sum_{i=1}^n S_{t_i}.$$

- Asian options are natural because:
 - procurement occurs continuously;
 - averaging reduces manipulation risk;
 - it better matches the operational exposure;
 - the premium is lower than for vanilla options.
- *Suggested image:* time series showing daily oil prices and the monthly average.

Asian Options: Numerical Methods

- Analytical approximations.
 - Geometric Asians admit closed forms.
 - Arithmetic Asians require approximation.
 - Moment-matching methods: Turnbull–Wakeman, Levy approximation.
- PDE methods.
 - Additional state variable for the running average;
 - dimension reduction from 2D to 1D via similarity transformation (Rogers & Shi, 1995).
- COS method.
 - Fourier-cosine expansion at each monitoring date (backward recursion);
 - the CF of the *arithmetic* average is not closed-form even under affine models – approximated via cumulants or via the one-step conditional CF.
 - *COS reference*: Zhang & Oosterlee, *SIAM J. Financial Math.* **4**(1), 399–426 (2013).
- Monte Carlo.
 - Most flexible method.
 - Natural for path dependence.
 - Variance reduction often needed.

Asian Options: References

Foundational pricing and approximation methods

- Kemna, A. G. Z., & Vorst, A. C. F. (1990) – “A Pricing Method for Options Based on Average Asset Values”, *J. Banking & Finance* **14**(1), 113–129. *Geometric Asian closed form; basis for control variates.*
- Turnbull, S. M., & Wakeman, L. M. (1991) – “A Quick Algorithm for Pricing European Average Options”, *J. Financial and Quantitative Analysis* **26**(3), 377–389. *Moment-matching approximation.*
- Levy, E. (1992) – “Pricing European Average Rate Currency Options”, *J. International Money and Finance* **11**(5), 474–491. *Lognormal approximation for arithmetic Asians.*
- Geman, H., & Yor, M. (1993) – “Bessel Processes, Asian Options, and Perpetuities”, *Math. Finance* **3**(4), 349–375. *Laplace-transform framework.*

PDE and dimension-reduction methods

- Rogers, L. C. G., & Shi, Z. (1995) – “The Value of an Asian Option”, *J. Applied Probability* **32**(4), 1077–1088. *Dimension-reduced PDE formulation.*
- Vecer, J. (2001) – “A New PDE Approach for Pricing Arithmetic Average Asian Options”, *J. Computational Finance* **4**(4), 105–113. *Unified one-dimensional PDE approach.*
- Fouque, J.-P., & Han, C.-H. (2003) – “Pricing Asian Options with Stochastic Volatility”. *Vecer-type reduction under mean-reverting SV.*
- Dubois, F., & Lelievre, T. (2005) – “Efficient Pricing of Asian Options by the PDE Approach”, *J. Computational Finance*. *Numerical PDE stabilisation and implementation.*

Transform, Fourier, and COS methods

- Fusai, G., & Abrahams, I. D. (2007) – “An Exact Analytical Solution for Discrete Arithmetic Asian Options”, *Finance and Stochastics* **11**, 349–375.
- Zhang, B., & Oosterlee, C. W. (2013) – “Efficient Pricing of European-Style Asian Options under Exponential Lévy”

Digital/Binary Options: Definition

- Digital options pay a fixed amount if a condition is met.
- *Cash-or-nothing call*: $\mathbf{1}_{\{S_T > K\}}$.
- *Asset-or-nothing call*: $S_T \mathbf{1}_{\{S_T > K\}}$.
- Key properties:
 - discontinuous payoff;
 - Dirac-like delta and sign-flipping gamma near the strike at expiry (*pin risk*);
 - building block of structured products;
 - linked to the risk-neutral distribution (CDF), via $C_{\text{digital}}(K) = e^{-rT} \mathbb{Q}(S_T > K) = -\partial C / \partial K$.

Digital Options: Real-World Example

- **FX corporate hedge.** A European exporter receives USD revenues and is exposed to a weak USD (i.e. a high EUR/USD rate).
- A digital option can pay a fixed amount only if

$$\text{EUR/USD} > 1.10 \quad \text{at maturity,}$$

offsetting the FX loss in that scenario.

- Applications:
 - structured deposits;
 - yield-enhancement notes;
 - interest-rate range accruals;
 - autocallables.
- *Suggested image:* payoff diagram showing the discontinuity at the strike.

Digital Options: Numerical Methods

- Closed-form pricing.
 - Black–Scholes closed forms exist.
 - Direct connection to vanilla option derivatives.
- PDE methods.
 - Requires careful handling of payoff discontinuity.
 - Grid smoothing often used.
- Fourier / COS methods.
 - Very efficient for discontinuous payoffs (the indicator payoff has closed-form cosine coefficients);
 - digitals are handled by the original European COS method as a special case.
 - *COS reference*: Fang & Oosterlee, *SIAM J. Sci. Comput.* **31**(2), 826–848 (2008).
- Monte Carlo.
 - Straightforward implementation.
 - Variance can be high near the strike.

- Core references:
 - Black & Scholes (1973);
 - Breeden & Litzenberger (1978);
 - Fang & Oosterlee (2008) – COS method.
- Key insight: digital prices are linked to the derivative of vanilla option prices,

$$\frac{\partial C}{\partial K}.$$

- This connects digital options to
 - implied distributions;
 - local-volatility extraction;
 - static replication.

Basket and Worst-of Options: Definition

- *Basket option* – depends on weighted assets:

$$\max\left(\sum_{i=1}^n w_i S_i(T) - K, 0\right).$$

- *Worst-of option* – depends on the worst-performing asset:

$$\max\left(\min_i S_i(T) - K, 0\right).$$

- *Market convention*: term sheets express the payoff in **performances** $S_i(T)/S_i(0)$ (not absolute levels), so that strikes are stated as percentages and assets at different absolute levels are comparable.
- **Key risk factor**: **correlation**.

Basket/Worst-of: Real-World Example

- **Worst-of autocallable.** Underlying basket:
 - EURO STOXX 50;
 - S&P 500;
 - Nikkei 225.
- Investor receives:
 - a 10% coupon if the worst performer is above the barrier;
 - potential early redemption;
 - downside linked to the worst-performing asset.
- **Why banks issue them:**
 - correlation risk premium;
 - retail demand for enhanced yield;
 - structured-product margins.
- **Convention:** the embedded option in a worst-of autocallable is in practice almost always a worst-of *put* (the investor is short downside on the weakest asset)².

² Investors buy the note for the high coupon. To fund that coupon the bank needs the investor to take on a risk worth selling – in practice the downside risk on the *worst-performing* index in the basket. Concretely, if at maturity the note has not autocalled and the worst index has fallen below a barrier (e.g. 60% of initial), the investor's redemption is reduced by the worst index's negative return. Economically this is a short position in a worst-of *put*: investor pays out when $\min_i S_i(T)/S_i(0)$ falls below the barrier. The textbook "call on the min" formula on the previous slide is mathematically clean but not the structure actually traded in retail markets.

Basket/Worst-of: Numerical Methods

- Main challenge: multi-dimensionality.
- Analytical approximations.
 - Moment matching;
 - lognormal approximations.
- PDE methods.
 - Computationally expensive in high dimensions;
 - curse of dimensionality.
- Fourier/COS methods.
 - Efficient under affine joint characteristic functions;
 - 2-asset basket / call-on-min: Ruijter & Oosterlee, *SIAM J. Sci. Comput.* **34**(5), B642–B671 (2012);
 - 3-asset spread: Pellegrino & Sabino (2016);
 - for worst-of / basket on ≥ 4 assets: no dedicated COS paper found – curse of dimensionality limits practical use.
- Monte Carlo.
 - Industry standard;
 - flexible for high dimensions;

Basket/Worst-of: References

Foundational multi-asset and rainbow options

- Margrabe, W. (1978) – “The Value of an Option to Exchange One Asset for Another”, *J. Finance* **33**(1), 177–186.
Exchange / spread option closed form.
- Stulz, R. (1982) – “Options on the Minimum or Maximum of Two Risky Assets”, *J. Financial Economics* **10**(2), 161–185.
Two-asset call/put on min/max – basis for worst-of.
- Johnson, H. (1987) – “Options on the Maximum or the Minimum of Several Assets”, *J. Financial and Quantitative Analysis* **22**(3), 277–283.
- Rubinstein, M. (1991) – “Somewhere Over the Rainbow”, *Risk* **4**(11), 63–66. *Practitioner overview of rainbow options.*

Basket-pricing approximations and bounds

- Gentle, D. (1993) – “Basket Weaving”, *Risk* **6**(6), 51–52. *Geometric-mean lognormal approximation.*
- Curran, M. (1994) – “Valuing Asian and Portfolio Options by Conditioning on the Geometric Mean Price”, *Management Science* **40**(12), 1705–1711.
- Milevsky, M. A., & Posner, S. E. (1998) – “A Closed-Form Approximation for Valuing Basket Options”, *J. Derivatives* **5**(4), 54–61. *Reciprocal-gamma approximation.*
- Beisser, J. (2001) – conditional-expectation bounds (PhD thesis, Mainz).
- Krekel, M., de Kock, J., Korn, R., & Man, T.-K. (2004) – “An Analysis of Pricing Methods for Basket Options”, *Wilmott Magazine*.

Transform, Fourier, and COS methods

- Hurd, T. R., & Zhou, Z. (2010) – “A Fourier Transform Method for Spread Option Pricing”, *SIAM J. Financial Math.* **1**, 142–157.
- Ruijter, M. J., & Oosterlee, C. W. (2012) – “Two-Dimensional Fourier Cosine Series Expansion Method for Pricing

Cliquet Options: Definition

- Cliquet options **lock in periodic returns**.
- Typical payoff:

$$\sum_{i=1}^n \min(\max(R_i, F), C),$$

where

- R_i – periodic return;
- F – floor;
- C – cap.
- Key characteristics:
 - path dependent;
 - periodic reset feature;
 - local-volatility sensitive;
 - popular in insurance products.

Cliquet Options: Real-World Example

- **Capital-protected structured note.** Investor receives:
 - principal protection at maturity;
 - annual equity upside participation;
 - gains locked in annually;
 - an annual cap of 8%.
- Popular in:
 - insurance products;
 - wealth management;
 - retail structured products.
- **Why investors like them:**
 - smooth return profile;
 - behavioral appeal;
 - downside protection.

Cliquet Options: Numerical Methods

- PDE methods.
 - Reset features create jump conditions;
 - state variable updated at reset dates.
- Fourier/COS methods.
 - Efficient for periodic structures (one-step CF per reset period);
 - **no dedicated COS paper found for the pure cliquet derivative;**
 - closest: COS applied to cliquet-style equity-indexed annuities (MPP/APP) – Deng, Dulaney, McCann & Yan, *Variance* (Casualty Actuarial Society) (2017);
 - alternative Fourier method for cliquets: PROJ / frame-duality (Kirkby & Nguyen) – not the COS method.
- Monte Carlo.
 - Widely used in practice;
 - flexible for local/stochastic-vol models.
- Tree/lattice methods.
 - Useful for pedagogical examples;
 - less practical in high dimensions.

- Core references:
 - Wilmott, *Paul Wilmott on Quantitative Finance* – cliquet chapter;
 - Bernard & Boyle (2008) – locally-capped contracts;
 - Hieber & Scherer (2012) – regime-switching cliquets.
- Practical modeling issues:
 - forward-skew exposure;
 - local-volatility calibration;
 - vega profile across reset dates;
 - smile dynamics matter critically.

Asian Options: Moment-Matching Procedure

- Approximate the arithmetic average by a lognormal variable:

$$A_T = \frac{1}{n} \sum_{i=1}^n S_{t_i}.$$

- Steps:
 - compute forward prices $F(0, t_i)$;
 - compute the first moment $m_1 = \mathbb{E}[A_T] = \frac{1}{n} \sum_{i=1}^n F(0, t_i)$;
 - compute the second moment $m_2 = \mathbb{E}[A_T^2] = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[S_{t_i} S_{t_j}]$;
 - fit lognormal parameters μ_A, σ_A from $\tilde{m}_1, m_2$;
 - price using a Black-style formula on $\tilde{A}_T$.
- **Examples:** Turnbull–Wakeman and Levy approximations.

Asian Options: PDE Procedure (constant volatility)

- Use an extra state variable for the running average: $V = V(t, S, A)$.
- Steps:
 - define the running average A_t ;
 - write the joint dynamics of (S_t, A_t) ;
 - derive the pricing PDE using no-arbitrage;
 - set the terminal condition $V(T, S, A) = \max(A - K, 0)$;
 - build grids for time, spot, and average;
 - step backward from maturity;
 - interpolate $V(0, S_0, A_0)$.
- **Trade-off:** accurate, but more expensive than a vanilla PDE because of the extra dimension.
- **Next slides:** extend this to *stochastic* volatility following Fouque & Han (2003), and reduce the spatial dimension by one via Vecer's trick.

Asian under Stochastic Volatility: Model Setup

- Reference: *Fouque & Han, "Pricing Asian Options with Stochastic Volatility" (2003)*.
- Risk-neutral dynamics (paper, eqs. (2)–(3)):

$$\begin{aligned}dS_t &= rS_t dt + f(Y_t) S_t dW_t^*, \\dY_t &= [\alpha(m - Y_t) - \beta \Lambda(Y_t)] dt + \beta(\rho dW_t^* + \sqrt{1 - \rho^2} dZ_t^*).\end{aligned}$$

- W^*, Z^* are independent $\mathbb{P}^*$ -Brownian motions; $\rho \in (-1, 1)$ encodes leverage; $\sigma_t = f(Y_t)$ with Y a mean-reverting (OU prototype) volatility factor; $\Lambda(\cdot)$ is the combined market price of risk.
- Path dependence: introduce the running integral

$$I_t = \int_0^t S_u du, \quad dI_t = S_t dt.$$

- Under $\mathbb{P}^*$, the triple (S_t, Y_t, I_t) is Markov.
- **Note.** The paper rescales by a small $\varepsilon = 1/\alpha$ (fast mean reversion) to set up a singular-perturbation expansion (§4); we keep the unscaled form here for clarity.

Three-Dimensional Pricing PDE (Fouque & Han, §3.1)

- Floating-strike Asian call price:

$$P(t, s, y, I) = \mathbb{E}^* \left[e^{-r(T-t)} \left(S_T - \frac{I_T}{T} \right)^+ \mid S_t = s, Y_t = y, I_t = I \right].$$

- By Feynman–Kac, P solves the **three-dimensional PDE** (paper eq. (5)):

$$\begin{aligned} \frac{\partial P}{\partial t} + \frac{1}{2} f(y)^2 s^2 \frac{\partial^2 P}{\partial s^2} + \rho \beta f(y) s \frac{\partial^2 P}{\partial s \partial y} + \frac{1}{2} \beta^2 \frac{\partial^2 P}{\partial y^2} \\ + r \left(s \frac{\partial P}{\partial s} - P \right) + (\alpha(m - y) - \beta \Lambda(y)) \frac{\partial P}{\partial y} + s \frac{\partial P}{\partial I} = 0, \end{aligned}$$

with terminal condition $P(T, s, y, I) = (s - I/T)^+$.

- The four blocks:**

- second-order spot/vol/cross-derivative diffusion – from (S, Y) quadratic variation, with the cross term carrying the leverage ρ ;
- first-order drifts in s and y – risk-neutral drift of S and risk-adjusted drift of Y ;
- the $s \partial P / \partial I$ term – I is a pure-drift state with $dI = S dt$;
- the $-rP$ – discounting.

- Issue:** three spatial dimensions $\Rightarrow$ expensive grids. This motivates dimension reduction.

Vecer's Dimension Reduction: Replicating the Running Average

- **Key idea** (Vecer 2001, generalised to SV in §3.2). Replicate $\frac{1}{T} \int_0^T S_u du$ by a *self-financing* portfolio in stock and bond, using a *time-deterministic* number of shares

$$q(t) = \frac{1 - e^{-r(T-t)}}{rT}.$$

- Self-financing wealth X_t (paper eq. (7)):

$$dX_t = q(t) dS_t + r(X_t - q(t)S_t)dt = rX_t dt + q(t)(dS_t - rS_t dt).$$

- Choose initial wealth $X_0 = q(0)S_0 - e^{-rT}K_2$. Integrating gives (eq. (10)):

$$X_T = \frac{1}{T} \int_0^T S_u du - K_2.$$

- **Unified payoff** (eq. (11)):

$$h\left(\frac{1}{T} \int_0^T S_u du - K_1 S_T - K_2\right) = h(X_T - K_1 S_T),$$

covering *fixed-strike* ($K_1 = 0$) and *floating-strike* ($K_2 = 0$) Asians; h is positively homogeneous, $h(xy) = xh(y)$ – which is the case for calls and puts.

- Trading the deterministic $q(t)$ collapses the path dependence in I_t into a single stochastic state X_t .

Vecer Reduction: Change of Numeraire and 2D PDE (§3.2)

- Switch to the stock numeraire: define $\Psi_t = X_t/S_t$. By Itô + Girsanov (eqs. (15)–(17)),

$$d\Psi_t = (q(t) - \Psi_t) f(Y_t) d\widetilde{W}_t^*,$$

with $\widetilde{W}_t^* = W_t^* - \int_0^t f(Y_s) ds$ a Brownian motion under the stock measure $\widetilde{\mathbb{P}}^*$. The Y -drift picks up an extra $\rho\beta f(Y)$ term under $\widetilde{\mathbb{P}}^*$.

- Homogeneity of $h \Rightarrow P(0, s, y) = s u(0, \psi_0, y)$ with $\psi_0 = \frac{1-e^{-rT}}{rT} - e^{-rT} K_2/s$.
- By Feynman–Kac, $u(t, \psi, y)$ solves the **two-dimensional PDE** (paper eq. (21)):

$$\begin{aligned} \frac{\partial u}{\partial t} + \frac{1}{2}(\psi - q(t))^2 f(y)^2 \frac{\partial^2 u}{\partial \psi^2} + \rho\beta(q(t) - \psi)f(y) \frac{\partial^2 u}{\partial \psi \partial y} + \frac{1}{2}\beta^2 \frac{\partial^2 u}{\partial y^2} \\ + (\alpha(m - y) - \beta(\Lambda(y) - \rho f(y))) \frac{\partial u}{\partial y} = 0, \end{aligned}$$

with terminal $u(T, \psi, y) = h(\psi - K_1)$.

- Pay-off:** 3D PDE $\rightarrow$ 2D PDE – one fewer spatial dimension for the same accuracy.
- Special case (constant vol, Vecer 2001):** drop Y and the PDE collapses to a *one-dimensional* equation in ψ – the celebrated Vecer 1D Asian PDE.
- Caveat.** This PDE prices the *fresh* Asian (at $t = 0$); for a *seasoned* Asian, use Asian put–call parity (paper §3.3).

Asian Options: Monte Carlo Procedure

- Steps:

- simulate M risk-neutral price paths,

$$S_{t_{j+1}} = S_{t_j} \exp\left((r - q - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} Z_j\right);$$

- compute the path average $A_T^{(m)} = \frac{1}{n} \sum_{j=1}^n S_{t_j}^{(m)}$;
 - compute the discounted payoff $X^{(m)} = e^{-rT} \max(A_T^{(m)} - K, 0)$;
 - average: $\hat{V}_0 = \frac{1}{M} \sum_{m=1}^M X^{(m)}$;
 - report a confidence interval using $s_X/\sqrt{M}$.
- Useful control variate: the geometric Asian option, which has a closed form.

Asian Options: COS / Fourier Procedure

- Fourier-cosine methods expand the density or payoff on a finite interval.
- The CF of the arithmetic average is *not* closed-form, even under affine models. Two strategies in practice:
 - approximate the CF of A_T (e.g. matching the first few cumulants);
 - or run **backward COS recursion** at the monitoring dates $t_1, \dots, t_n$ using the conditional one-step CF (Zhang & Oosterlee, 2013).

- Steps (approximation route):

- approximate the CF φ_{A_T} of the average;
- choose the truncation interval $[a, b]$ using cumulants;
- compute the cosine payoff coefficients H_k ;
- evaluate

$$V_0 = e^{-rT} \sum_{k=0}^{N-1} \text{Re} \left[\varphi_{A_T} \left(\frac{k\pi}{b-a} \right) e^{-ik\pi a/(b-a)} \right] H_k;$$

- increase N and widen $[a, b]$ until stable.
- **Best use:** when a sufficiently accurate CF approximation is available.

Digital Options: Closed-Form Procedure

- Cash-or-nothing digital call: $\mathbf{1}_{\{S_T > K\}}$.

- Steps:

- compute

$$d_2 = \frac{\ln(S_0/K) + (r - q - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}};$$

- interpret $N(d_2)$ as the risk-neutral exercise probability;
- discount the fixed payoff, $V_0 = e^{-rT} N(d_2)$;
- for a payoff amount Q , $V_0 = Q e^{-rT} N(d_2)$.
- **Main intuition:** a digital prices a probability under the risk-neutral measure.

Digital Options: Static Replication Procedure

- Approximate a digital call by a tight **centred call spread**:

$$\mathbf{1}_{\{S_T > K\}} \approx \frac{(S_T - K + \frac{\varepsilon}{2})^+ - (S_T - K - \frac{\varepsilon}{2})^+}{\varepsilon}.$$

- Steps:
 - choose a small strike spacing ε ;
 - buy $1/\varepsilon$ calls struck at $K - \frac{\varepsilon}{2}$;
 - sell $1/\varepsilon$ calls struck at $K + \frac{\varepsilon}{2}$;
 - price the spread using vanilla implied volatility;
 - let ε shrink to approximate the digital.
- Centring around K gives second-order accuracy in ε ; a one-sided spread gives only first-order.
- Risk-neutral link:

$$\frac{\partial C}{\partial K} = -e^{-rT} \mathbb{Q}(S_T > K).$$

- **Teaching point:** digitals connect option prices to the risk-neutral distribution.

Digital Options: PDE and Monte Carlo Procedure

- PDE.

- Set the terminal condition $V(T, S) = \mathbf{1}_{\{S > K\}}$;
- build time and spot grids;
- align or smooth the discontinuity near K ;
- step backward using implicit or Crank–Nicolson schemes;
- interpolate to obtain $V(0, S_0)$.

- Monte Carlo.

- Simulate terminal values $S_T^{(m)}$;
- evaluate $X^{(m)} = e^{-rT} \mathbf{1}_{\{S_T^{(m)} > K\}}$;
- average the discounted payoffs;
- use conditional Monte Carlo or smoothing to reduce variance.

- Numerical issue: the discontinuity creates noisy Greeks.

Basket Options: Moment-Matching Procedure

- Basket variable: $B_T = \sum_{i=1}^n w_i S_i(T)$.
- Steps:
 - compute the expected basket value, $m_1 = \mathbb{E}[B_T] = \sum_{i=1}^n w_i F_i(0, T)$;
 - compute the second moment using volatilities and correlations,
 $m_2 = \sum_i \sum_j w_i w_j \mathbb{E}[S_i(T) S_j(T)]$;
 - fit a lognormal distribution to B_T ;
 - price the basket call using a Black-style formula.
- **Main sensitivity:** the basket variance depends strongly on correlation,

$$\text{Var}(B_T) = \sum_i \sum_j w_i w_j \text{Cov}(S_i, S_j).$$

Basket/Worst-of Options: Monte Carlo Procedure

- **Industry standard** for multi-asset exotics.
- Steps:
 - estimate the volatility vector and correlation matrix ρ ;
 - compute the Cholesky factor $\rho = LL^\top$;
 - generate independent standard normals Z ;
 - create correlated shocks $\tilde{Z} = LZ$;
 - simulate each asset path under the risk-neutral measure;
 - compute the payoff (worst-of call):

$$X^{(m)} = e^{-rT} \max\left(\min_i S_i^{(m)}(T) - K, 0\right);$$

- average across paths and compute the standard error.
- **Enhancements:** antithetic sampling, quasi-Monte Carlo, control variates.

Basket/Worst-of Options: PDE and Fourier Procedure

- PDE procedure.

- Define $V(t, S_1, \dots, S_n)$;
- include cross-derivative terms from correlation,

$$\rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j};$$

- apply the terminal basket or worst-of payoff;
- step backward on a multi-dimensional grid.

- Fourier / COS procedure.

- Obtain the joint characteristic function of log-prices;
- choose a multidimensional truncation region;
- compute the payoff coefficients;
- sum the multidimensional expansion.

- Limitation: both PDE and COS become expensive as dimension grows.

Cliquet Options: Decomposition Procedure

- The cliquet payoff is built from capped/floored periodic returns:

$$R_i = \frac{S_{t_i}}{S_{t_{i-1}}} - 1, \quad \ell_i = \min(\max(R_i, F), C), \quad \text{Payoff} = \sum_{i=1}^n \ell_i.$$

- Steps:
 - split the product into reset periods;
 - compute each local return R_i ;
 - apply the local floor F and cap C ;
 - accumulate local returns;
 - apply any global cap/floor at maturity;
 - discount the expected payoff.
- **Important:** global caps/floors make the product path dependent in accumulated return.

Cliquet Options: Monte Carlo Procedure

- Steps:
 - simulate paths at reset dates $t_0, t_1, \dots, t_n$;
 - for each period, $R_i^{(m)} = S_{t_i}^{(m)} / S_{t_{i-1}}^{(m)} - 1$;
 - apply the local cap/floor, $\ell_i^{(m)} = \min(\max(R_i^{(m)}, F), C)$;
 - aggregate, $G^{(m)} = \sum_{i=1}^n \ell_i^{(m)}$;
 - apply the global cap/floor if present;
 - discount and average,

$$\hat{V}_0 = \frac{1}{M} \sum_{m=1}^M e^{-rT} G^{(m)}.$$

- **Practical use:** easy to combine with local-volatility, stochastic-volatility, and hybrid models.

Cliquet Options: PDE and COS Procedure

- PDE.
 - Use state variables $V(t, S, G)$, where G is the accumulated return;
 - solve the PDE between reset dates;
 - at each reset date, update $G_i = G_{i-1} + \min(\max(R_i, F), C)$;
 - continue backward through all reset dates.
- COS / Fourier.
 - Express the payoff in terms of log-return over each reset period;
 - use the characteristic function of period returns;
 - expand the capped/floored return payoff in a cosine series;
 - recursively aggregate the reset periods.
- Challenge: accumulated global caps/floors add another state variable.

Key Takeaways

- Exotic options exist to match real-world financial exposures.
- Post-2008 markets shifted toward simpler and more hedgeable products.
- Asian, digital, basket/worst-of, and cliquet products remain highly relevant.
- Numerical methods are central because closed forms are often unavailable.
- Monte Carlo remains dominant in industry implementations.
- Modern themes in exotic-option markets:
 - correlation risk;
 - volatility smile dynamics;
 - XVA and funding adjustments;
 - regulatory capital constraints;
 - structured-product demand.

- Books:

- J. Hull, *Options, Futures and Other Derivatives*;
- P. G. Zhang, *Exotic Options*;
- A. Lipton, *Mathematical Methods for Foreign Exchange*;
- R. Rebonato, *Volatility and Correlation*.

- Important papers:

- F. Black and M. Scholes (1973);
- D. T. Breeden and R. H. Litzenberger (1978);
- S. M. Turnbull and L. M. Wakeman (1991);
- J. Vecer (2001) – PDE dimension reduction for arithmetic Asian options;
- J.-P. Fouque and C.-H. Han (2003) – “Pricing Asian Options with Stochastic Volatility”;
- F. Fang and C. W. Oosterlee (2008).

Questions?